

Parameter	Symbol	Formula	Calculation	Estimated Value	Unit	Description
Frontal Area	A	$A = W \times H \times k$	-	-	m ²	
Drag Coefficient	Cd	SEM Regulation	-	70	kg	
Drag Force	Fd	$1/2 \cdot \rho \cdot v^2 \cdot A \cdot C_d$			kg	
Length	L		-			
Height	H				-	
Front Track Width		-	-	1100	mm	Estimated within SEM regulations and Engineering Assumption
Rear Track Width		-	-	850	mm	
Wheel Base		-	-	1600	mm	
Air Density	ρ	Standard atmosphere	-	1.225	kg/m ³	
Turning Radius	Tr	-	-	6	m	Estimated within SEM regulations
Traction	Ft	$F_{rr} + F_d + W \cos$	-	-	-	
Velocity	Vavg	s/t	$((14,8 \times 1000 \times 4) - 24) / (35 \times 60)$	28	m/s	Estimated by Final Project rules
Rolling Resistance Force	Fr	$F_r = C_{rr} \times m \times g$	$0.004 \times 155 \times 9.81$	6.08	N	Tyre rolling resistance force